WORKBOOK REPORT · SHEET “STD AND CV”

Standard deviation and coefficient of variation

Pizza price example · 10 pizza prices in New York, in dollars and in Mexican pesos (1 $ = 18.81 MXN)

Data set

#	NY Dollars	Pesos
1	$ 1.00	MXN 18.81
2	$ 2.00	MXN 37.62
3	$ 3.00	MXN 56.43
4	$ 3.00	MXN 56.43
5	$ 5.00	MXN 94.05
6	$ 6.00	MXN 112.86
7	$ 7.00	MXN 131.67
8	$ 8.00	MXN 150.48
9	$ 9.00	MXN 169.29
10	$ 11.00	MXN 206.91

Unchanged from the workbook. Pesos = dollars × 18.81 (formula =C5*18.81 in cells D5:D14).

Results

Measure	Dollars	Pesos
Mean	$ 5.50	MXN 103.46
Sample variance	$² 10.72	MXN² 3793.69
Sample standard deviation	$ 3.27	MXN 61.59
Coefficient of variation	**0.60**	**0.60**

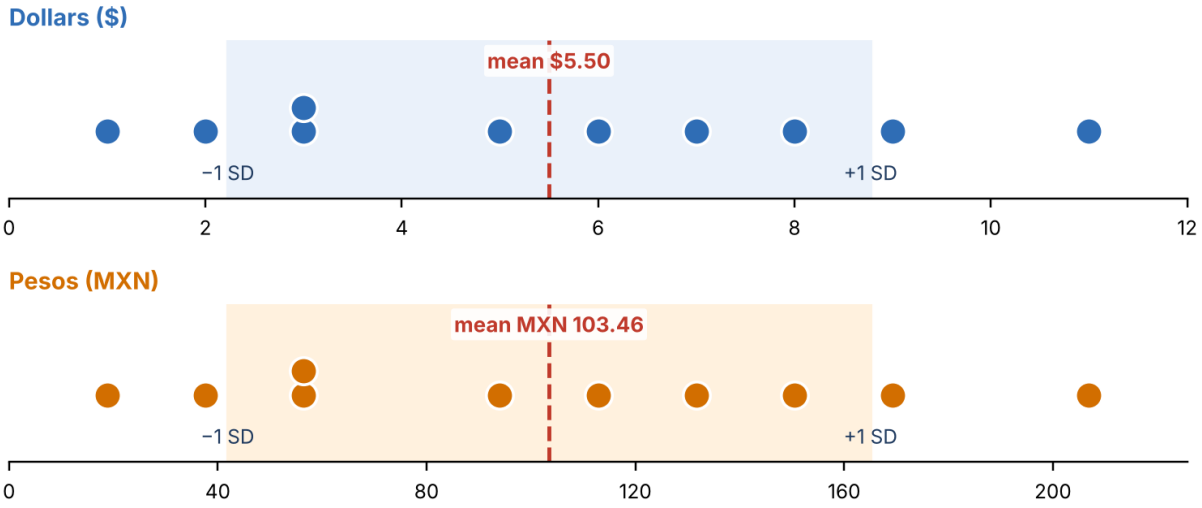
How each result is calculated

Measure	Excel formula
Mean	=AVERAGE(C5:C14)
Sample variance	=VAR.S(C5:C14)
Sample SD	=STDEV.S(C5:C14)
Sample CV	=G7/G5

Dollar column shown; the pesos column uses D5:D14.

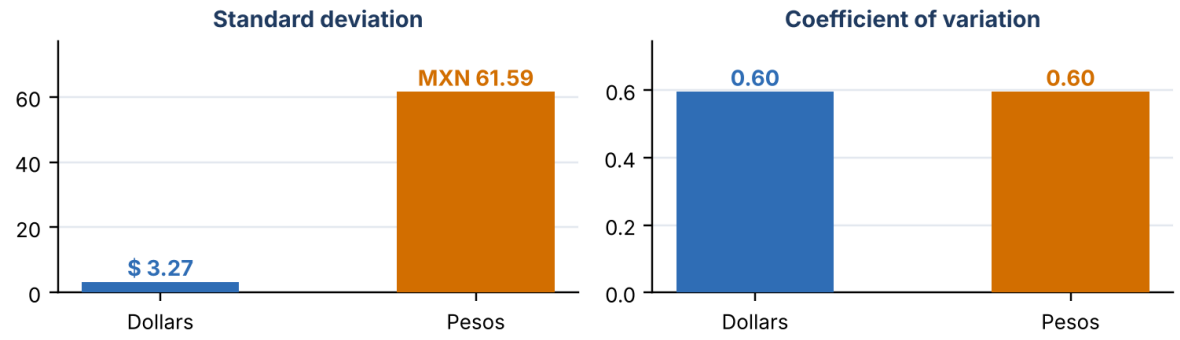
Key finding
The standard deviation looks very different in the two currencies ($ 3.27 vs MXN 61.59), but the **coefficient of variation is 0.60 in both**. The prices are the same pizzas, so their relative variability must be the same, and the CV shows exactly that.

Same shape, different scale



Each dot is one price. The red line is the mean and the shaded band is one standard deviation either side of it. The dollar and peso charts have the same layout because every peso price is the dollar price × 18.81.

Standard deviation vs. coefficient of variation



Step-by-step calculation (sample, n = 10, so divide by n − 1 = 9)

#	Dollars x	x − 5.50	(x − 5.50)²	Pesos x	x − 103.455	(x − 103.455)²
1	1.00	-4.50	20.25	18.81	-84.645	7,164.78
2	2.00	-3.50	12.25	37.62	-65.835	4,334.25
3	3.00	-2.50	6.25	56.43	-47.025	2,211.35
4	3.00	-2.50	6.25	56.43	-47.025	2,211.35
5	5.00	-0.50	0.25	94.05	-9.405	88.45
6	6.00	+0.50	0.25	112.86	+9.405	88.45
7	7.00	+1.50	2.25	131.67	+28.215	796.09
8	8.00	+2.50	6.25	150.48	+47.025	2,211.35
9	9.00	+3.50	12.25	169.29	+65.835	4,334.25
10	11.00	+5.50	30.25	206.91	+103.455	10,702.94
Σ	**55.00**	**0.00**	**96.50**	**1034.55**	**0.00**	**34,143.25**

Step	Dollars	Pesos
1. Mean = Σx ÷ n	55.00 ÷ 10 = $ 5.50	1034.55 ÷ 10 = MXN 103.46
2. Variance s^2 = Σ(x − mean)² ÷ (n − 1)	96.50 ÷ 9 = 10.72	34,143.25 ÷ 9 = 3793.69
3. SD s = √s^2	√10.72 = 3.27	√3793.69 = 61.59
4. CV = s ÷ mean	3.27 ÷ 5.50 = 0.60	61.59 ÷ 103.46 = 0.60

Why the CV does not change

Quantity	Dollars → Pesos	Factor	Check
Mean	5.50 → 103.455	× 18.81	5.50 × 18.81 = 103.455
Standard deviation	3.27 → 61.59	× 18.81	3.27 × 18.81 = 61.59
Variance	10.72 → 3,793.69	× 18.81^2 = 353.82	10.72 × 353.82 = 3793.69
CV	**0.60 → 0.60**	**× 1 (unchanged)**	18.81 cancels in SD ÷ mean

About this report
The data set (the ten dollar prices and the ten peso prices) and the exchange rate are reproduced exactly as they appear in the workbook, and all workbook results (mean, SD, CV) match to the cell. Two layout changes only: the sample variances, which the workbook stores as typed text ("$² 10.72" and "MXN² 3793.69"), are shown as ordinary numbers and recalculated with =VAR.S; the results agree with the typed values.